Qi Huang

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EDUCATION

The University of Chicago **Sept 2025 – Jun 2027**
M.Sc. in Computational and Applied Mathematics *Chicago, US*

The University of Liverpool **Sept 2023 – Jun 2025**
B.Sc. in Mathematics, First Class Honours · Ranked Top 1% in Major *Liverpool, UK*

Xi'an Jiaotong-Liverpool University **Sept 2021 – Jul 2023**
B.Sc. in Applied Mathematics *Suzhou, China*

RESEARCH EXPERIENCE

Causal Inference

Sensitivity Analysis for Matched Observational Studies **Nov 2025 – Present**
University of Chicago, Department of Statistics | *Mentor:* ***Prof. Xinran Li***

- Developed a covariate-adaptive studentized sensitivity analysis for Neyman's weak null under heterogeneous treatment effects in matched pairs and full matching, using bounded perturbations of baseline propensity odds.
- Established mean control and sharpness within a specified score class, conditional on the within-set assignment law; estimated propensities require endpoint-coverage or plug-in rate conditions.
- Derived design sensitivity under a specified favorable model and optimized balanced comparison weights for full matching.

LLMs and Machine Learning

Scientific World Models and LLM Agents **Mar 2026 – Present**
Princeton University, Zhuang Lab | *Mentor:* ***Prof. Zhuang Liu***

- Developing research-decision support that combines research context with retrieved literature and experimental records to compare candidate actions, allowing ties or abstention when evidence is insufficient.
- Exploring integration into autonomous research workflows to guide experiment selection and branch expansion, with evaluation against observed experimental outcomes.

AI Breakthroughs and Intellectual Diversity **Sept 2025 – Jul 2026**
University of Chicago, Knowledge Lab | *Mentor:* ***Prof. James A. Evans***

- Co-designed and implemented agent-based simulations examining how reciprocal peer-review policies can concentrate accepted topics and reduce diversity among accepted authors.
- Co-authored a study of 7,180,635 computer-science papers, 73,089 peer reviews, and topic shifts among 3,707,765 researchers across 36 years; submitted to Nature Machine Intelligence.

Audio-Visual Test-Time Adaptation **Sept 2025 – Jun 2026**
University of Wisconsin–Madison | *Mentor:* ***Prof. Xiao Luo***

- Developed HEART: two complementary half-routes mix source and adapted audio–visual features to check anchor pseudo-labels; a batch gate uses support coverage and contrast to weight the adaptation loss.
- Recovers Self-Anchor when batch evidence is uninformative. On AVRobustBench at severity 5, mean accuracy rises from 46.98% to 48.90% on VGGSound-2C and remains similar on KS32-2C (65.73% to 65.75%).

PUBLICATIONS AND MANUSCRIPTS

Causal Inference

- **Qi Huang***, Pengyun Wang*, Xinran Li. *Covariate-Adaptive Studentized Sensitivity Analysis for Neyman's Weak Null in Matched Designs*. Working paper (*equal contribution)

LLMs and Machine Learning

- **Qi Huang**, Xiao Luo. *Half-routing Consistency Guidance with Batch-aware Gating for Audio-Visual Test-Time Adaptation*. Manuscript (HEART)
- Yuxuan Hu, Yushan Pan, Minghui Zhu, Xiaojiang He, Nan Xiang, **Qi Huang**, Ruizhi Ning, Yanlong Cao, Zhijie Xu. *QEGF: A Quantum-Inspired Geometric and Energy-Based Fusion Framework for Multimodal Sentiment Analysis*. Accepted, IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2026
- Pengyun Wang, Junyu Luo, Yanxin Shen, Ming Zhang, Shaoen Qin, Hanwen Xing, Siyu Heng, **Qi Huang**, Xiao Luo. *A Comprehensive Graph Pooling Benchmark: Effectiveness, Robustness and Generalizability*. Major revision, ACM Transactions on Knowledge Discovery from Data (TKDD)
- Jia Huang, Hongyou Zhou, **Qi Huang**, Hongbin Liu, Peng Zhang, Zhengqi Sun, Lumin Xu. *PVI-DPO: Perceptual Video Inpainting via Informative-Safe Preference Mining and Region-Aware Direct Preference Optimization*. Submitted to AAAI Conference on Artificial Intelligence (AAAI) 2027
- Beichen Lu, Honglin Bao, **Qi Huang**, Qingguo Meng, James A. Evans. *Stronger Artificial Intelligence Breakthroughs Hasten Their Own Exhaustion*. Submitted to Nature Machine Intelligence

INTERNSHIPS

iFLYTEK **Jun 2025 – Sept 2025**
Machine Learning Research Intern *Hefei, China*

- Trained and evaluated Transformer baselines in PyTorch on in-house text corpora; improved BLEU by +2.3 (27.6 → 29.9) through regularization and decoding tuning.
- Trained Conformer baselines on internal speech data; reduced WER from 10.3% to 8.9% via LR scheduling, SpecAugment, and label smoothing.

Alibaba Group **Jul 2024 – Aug 2024**
Data Analyst Intern *Hangzhou, China*

- Cleaned and analyzed consumer data using AARRR metrics; supported retention analysis and precision marketing with RFM-based customer segmentation.
- Built activity and retention visualizations in Matplotlib and Pandas to support management decision-making.

ACADEMIC SERVICE

Reviewer: ICLR 2027; AAAI 2027; Forecast@ICML 2026 Workshop (Forecasting as a New Frontier of Intelligence)

SKILLS

Programming: Python, SQL, R, MATLAB, Java, VBA, Maple
Tools: PyTorch, Pandas, Matplotlib, Tableau, Power BI, Jupyter Notebook, LaTeX, Excel